FlashSkip: Latency-Guided Structured Pruning for Transformer Encoders

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Abstract

Large transformer models deliver state-of-the-art accuracy but carry a prohibitive inference cost. We present FlashSkip, a structured pruning framework that removes redundant attention heads and feed-forward channels jointly under a learned latency budget.

FlashSkip reaches 91.2% accuracy on the GLUE benchmark while cutting inference latency by 38% relative to the dense baseline, and it requires no retraining of the remaining weights.

Introduction

Deploying twelve-layer encoders on edge hardware remains difficult because memory bandwidth, not raw compute, dominates end-to-end latency. Previous pruning work removes weights in a single pass and then fine-tunes for days on expensive accelerators.

Our approach differs in two ways. First, a differentiable latency predictor guides the pruning search directly, so the optimization target is the deployment metric itself. Second, we freeze all surviving weights and only calibrate biases, which makes the procedure practical on a single consumer GPU.

Methodology

We evaluate on the GLUE benchmark using the standard train and development splits, and we additionally profile latency on an NVIDIA Jetson Orin device at batch size one. The search runs for 48 GPU-hours on one A100 card, which is roughly six times cheaper than comparable pruning pipelines.

The pruning objective combines the cross-entropy loss with a learned gate over heads and channels. After pruning, a 2,000-step bias calibration recovers most of the lost accuracy without touching the pretrained matrices.

Results

FlashSkip achieves 91.2% average GLUE accuracy, within 0.4 points of the dense teacher, while the latency drops from 21.4 ms to 13.3 ms per sentence. Head pruning alone contributed 61% of the total speed-up, and the channel gates contributed the remaining 39%.

Ablations show that removing the latency predictor costs 1.1 points of accuracy at equal latency, confirming that the learned budget is the central component of the framework.

Limitations

Our evaluation is limited to encoder-only models and a single hardware target, so the behaviour on decoder architectures remains untested. The bias-calibration step also assumes calibration data from the same distribution as pretraining, which may not hold in domain-shifted deployments.